Chemistry

Unit 1 Review Answers

Section one—Multiple-choice questions

Question 1

D The group number is dependent on the number of valence electrons, i.e. the number of electrons in the outer electron shell.

Question 2

D Mass number is the sum of the number of protons and neutrons. For this ion the mass number is 11 + 12 = 23. The ion has one fewer electron than protons so it has a charge of +1.

Question 3

A The electronegativity increases up a group and increases across a period.

Question 4

C Elements in the same group of the periodic table have similar properties as they have the same number of valence electrons. Only N and P belong to the same group.

Question 5

B The reactivity of metals increases down a group and decreases across a period as the ionisation energy decreases.

Question 6

B A full outer shell of electrons is a stable arrangement. Atoms that already have complete (full) valence shells will therefore be unlikely to react with other atoms.

Question 7

C All atoms of the same element have the same number of protons in their nuclei.

Question 8

B All atoms of copper have the same atomic number (same number of protons). However, isotopes differ in the number of neutrons in the nucleus.

number of neutrons = mass number – atomic number.

So an atom of copper could have either 34 (63 – 29) or 36 (65 – 29) neutrons.

Question 9

C Isotopes have the same chemical properties (because the electron configurations are the same) but different physical properties (because the masses of the atoms are different).

Question 10

B Isotopes have the same atomic number but different mass numbers.

Question 11

C Y has an atomic number of 20, so it has 20 protons in the nucleus. Its electronic configuration is 2,8,8 so it has 18 electrons. Because it has two more protons than electrons it will have a positive charge.

Question 12

D Electrons gain energy which promotes them to a higher energy level. The absorption of this energy and can be observed as lines of darkness at particular frequencies on a spectrum.

Question 13

B Distillation requires a variation in boiling point between substances being separated.

Question 14

D A heterogeneous mixture has properties and composition that vary throughout the mixture.

Question 15

D Dilute hydrochloric acid is an aqueous solution, air is a mixture of gases.

Question 16

D In an atom there are forces of attraction between the positively charged nucleus/protons and the negatively charged electrons. There are also forces of repulsion between positively charged protons in the nucleus and between negatively charged electrons surrounding the nucleus.

Question 17

D Magnesium chloride is an ionic substance which will not conduct electricity unless the ions are free to move (molten or in solution).

Question 18

C The charge on a phosphate ion is 3– and on a magnesium ion is 2+, so the correct formula is $Mg_3(PO_4)_2$.

Question 19

D Element X would have four valence electrons so needs four electrons to achieve a stable outer shell configuration. Element Y has six valence electrons, so needs two to achieve stability. Hence, one X atom would bond with two Y atoms.

Question 20

B The formation of an ionic bond requires one of the atoms to gain electrons to fill the outer shell and one atom to lose electrons to leave a full outer (valence) shell of electrons.

Question 21

C Ionic bonding involves the transfer (not sharing) of electrons, normally from a metal to a non-metal.

Question 22

C The electrostatic forces of attraction between particles will determine the melting point of a solid. Na_2O is the only ionic substance listed, so it will have the strongest bonds (electrostatic attractions) between the particles.

Question 23

D In general, ionic compounds have a high melting temperature and covalent molecular compounds have a lower melting temperature. However, there are exceptions. For example, in diamond, the carbon atoms are held together by covalent bonds and diamond has a very high melting temperature. The same is true for hardness and brittleness.

In ionic compounds in the solid state, the ions are held in fixed position and so cannot carry a charge. However, in the molten state the ions are free to move so the compound is an electrical conductor in that state. Covalently bonded compounds have no charged particles and so are non-conductors.

Question 24

B The bond is created by the fact that the pair of (negative) electrons are attracted to the (positive) nuclei of two neighbouring atoms.

Question 25

C The structure of HCN is H–C≡N:

There is a triple bond between the C and N. Therefore, six electrons are shared between these two atoms.

Question 26

C The conduction of an electric current requires charged particles (ions or electrons) that are free to move. There are no ions formed in covalent compounds and the electrons are held in place in the covalent bonds.

Question 27

C SiO_2 (silica) is the only one that is a covalent network substance and therefore has strong covalent bonds throughout the structure. These bonds require a large amount of energy to break.

Question 28

C The relatively high surface area means that the nanomaterials can react or behave in different ways as there is more 'space' for molecular or atomic interactions.

Question 29

B The general formula for alkanes is C_nH_{2n+2}

Question 30

A It is an alkene with four carbons in the longest chain. For naming, the carbon chain should be numbered from the end closest to the branched group.

Question 31

C Only unsaturated organic compounds undergo addition reactions. Methylbutane is saturated.

Question 32

C Complete combustion produces carbon dioxide and water.

Question 33

C Using $n = m \times M$, $M = \frac{n}{m} = \frac{16.0}{0.235} = 68$ g

So the molar mass of the compound is 68 g mol^{-1}. The only compound with this molar mass is C_5H_8

Question 34

D The molecular formula for the compound is C_5H_{12}. Each molecule of pentane has 17 atoms. 4 moles of C_5H_{12} contains $4 \times 17 \times 6.02 \times 10^{23} = 4.09 \times 10^{25}$ atoms.

Question 35

C %H in $H_2O = \left(\frac{2 \times 1.0}{18.0}\right) \times 100$

$= 11\%$

Question 36

A *m*(urea) in 125 g fertiliser $= \frac{58.2}{100} \times 125 = 73.1$ g

%N in urea $= \frac{M(\text{N})}{M(\text{urea})} \times 100 = \frac{24 \times 100}{60} = 40.0\%$ nitrogen

So *m*(N) in 125 g fertiliser $= \frac{40.0}{100} \times 73.1\text{g} = 29.2$ g

Question 37

A 1 mole of CO_2 has a mass of 44 g ($12 + 2 \times 16 = 44$ g).

So the mass of 2 moles of CO_2 is $2 \times 44 = 88$ g

Question 38

B 1 mole of NaOH contains 2 moles of ions (1 mole of Na^+ ions and 1 mole OH^- ions)

1 mole of K_3PO_4 contains 4 moles of ions (3 moles of K^+ ions and 1 mole of PO_4^{3-} ions).

So 0.5 mole of K_3PO_4 contains the same number of ions as 1 mole of NaOH.

Question 39

D Crystallisation involves the formation of chemical bonds. The solid crystal will have less stored chemical energy (enthalpy) than the ions or particles dissolved in the solution.

Question 40

B The enthalpy of the chemicals decreases during the reaction. This energy is released to the environment, mainly as heat but also as some sound and kinetic energy.

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Section two—Short and extended answer questions

Question 1

a. When atoms absorb or emit energy, a spectrum is obtained.

Emission and absorption spectra are not continuous but contain discrete lines corresponding to discrete amounts of energy emitted or absorbed as electrons move from one energy level to another.

Bohr proposed that electrons can only move in fixed orbits or shells of particular energy and could only jump from one orbit to another without stopping in between, so only certain energies can be emitted or absorbed.

b. It does not explain why electrons move in circular orbits.

It does not explain why shells have particular energies. The mathematical model does not hold well for larger atoms.

Question 2

a. mass spectrometer

b. $A_r(\text{Ir}) = \dfrac{(37.30 \times 190.97) + (62.70 \times 192.97)}{100}$

$= 192$ (to 3 significant figures)

Question 3

Isotopes have the same number of protons and electrons but a different number of neutrons.

Therefore, the mass of different isotopes is different. This affects physical properties such as melting point and boiling point. If an atom has a larger mass, it will require more heat energy to increase its kinetic energy and separate it from other atoms. This is why the melting and boiling point of the heavier isotopes will be higher than he lighter isotopes.

However, chemical properties, such as types of reactions and reactivity, are dependent on the arrangement of electrons in the atom. As the number and arrangement of electrons does not vary between isotopes of the same element, the chemical properties will not vary.

Question 4

a. Vaporisation $\rightarrow$ Ionisation $\rightarrow$ Acceleration $\rightarrow$ Deflection $\rightarrow$ Detection

b. Vaporisation: a sample is heated to turn it into a gas.

Ionisation: the particles in the sample are converted into positive ions.

Acceleration: the charged ions are accelerated in an electric field (attracted to a negatively charged plate).

Deflection: the beam of ions is passed through a magnetic field that causes the ions to be deflected off their path.

Detection: ions of different masses are detected at different times/places.

Question 5

a. Chlorine is on the right side of the table and sodium is on the left. Atomic radius decreases across a period because the increasing nuclear charge pulls the outer-shell electrons more tightly to the nucleus, causing the volume of the atom to decrease.

b. Fluorine is further to the right on the periodic table than lithium and from left to right across the periodic table the nuclear charge increases. As nuclear charge increases, the electrons are held more tightly to the nucleus and more energy is required to remove the first one.

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c. Ba and Be are in the same group, with Be higher than Ba. The atom size increases down a group, meaning the outer electrons are further from the nucleus. Therefore, the outer electron on Be is held more tightly than on Ba and is less readily released.

Question 6

a. calcium and magnesium; same number of valence electrons (two)

b. neon; complete outer (valence) shell of electrons

c. potassium; relatively low nuclear charge and four electron shells/low core nuclear charge

d. Al_2S_3; aluminium ions have charge/valency of 3+ and sulfide ions have charge/valency of 2–

Question 7

a. **i.** Ca

ii. Ar

iii. C or Si

iv. Na

v. Li

vi. N

vii. F

b. Chemical reactivity increases because the outer shell electrons are further from the nucleus (and shielded from the nucleus by more inner electron shells) as one moves down the group and so less energy is required to remove them when the atom reacts.

Question 8

a. the total number of protons and neutrons in an atom

b. the total number of protons in an atom

c. the mass of an atom of the isotope relative to the mass of an atom of C-12, taken as 12 units exactly

d. 6.02×10^{23} (which is the number of particles in a mole)

e. an atom, or a group of atoms, that has either gained or lost one or more electrons and so carries an electrostatic charge

Question 9

a. The radiation absorbed corresponds to specific differences between the energy levels of the electrons in the atoms. Different elements will have different energy levels so the wavelength of the radiation has to be chosen to match the different gaps between the energy levels for that particular atom.

b. In atomic emission, the atom is given energy by an external source, normally heat. This excites electrons in the atoms, which can result in them moving to a higher energy level. As they move back to the ground state (lowest possible energy level) the energy that they lose is emitted as radiation, often in the form of visible light.

In atomic absorption, atoms are exposed to radiation and the electrons absorb the radiation of specific wavelengths. Instead of light being emitted, spectra will show black lines, which represent the radiation that has been absorbed by the atoms.

Question 10

A homogeneous mixture is a mixture that has uniform composition and properties throughout. Examples include salt water, apple juice, any solution and any alloy.

Question 11

a. K_3PO_4

b. Al_2O_3

c. $NaNO_3$

d. Fe_2S_3

Question 12

a. calcium nitrate

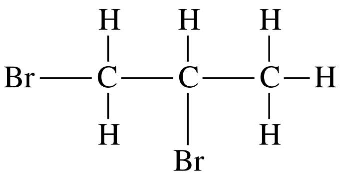
b. copper(II) carbonate

c. magnesium hydroxide

d. chromium(III) oxide

Question 13

a.



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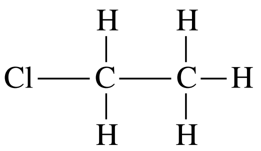

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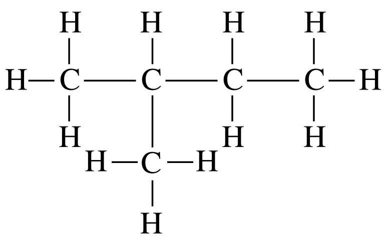


Question 14

a. 2,8,2

b. **i.** 1 pm = 10^{-12} m and 1 nm = 10^{-9} m, so 160 pm = 0.160 nm

ii. The sodium atom would have a larger radius because there are fewer protons in the nucleus, so the attraction for the outer shell electrons would be weaker, pulling them less strongly toward the nucleus.

c. **i.** A lattice of positively charged magnesium ions surrounded by a 'sea' of valence electrons. The lattice is held together by the electrostatic attraction between electrons and cations.

ii. The electrons are not localised (they are free to move) and so can conduct an electric current.

iii. The model cannot explain the differences in melting points, densities, electrical conductivities or magnetism between metals. (Any one of these is a correct answer.)

d. **i.** You would observe bubbles of gas being evolved, or an increase in temperature, or the magnesium dissolving.

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ *or*

$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$

ii. any suitable example such as K, Na, Ca

e. An alloy is a substance formed when other materials, usually carbon or other metals, are mixed with a metal.

Alloying increases the hardness or strength of a metal. (Resistance to corrosion is also an acceptable answer.)

Question 15

a. One mark for correct formula, one mark for appropriate use.

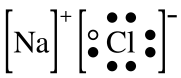
e.g. NaCl used for preserving food, $NaHCO_3$ as baking soda, NaF in toothpaste to harden tooth enamel

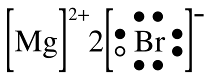
b. **i.** Though charged particles are present, they are held in fixed positions in the lattice so cannot move to carry a current.

ii. If a strong force is applied to a crystal of the compound, the layers of ions will move relative to one another causing ions of like charge to be adjacent and hence repel. The crystal thus shatters.

iii. A large amount of energy is needed to overcome the electrostatic attraction between oppositely charged ions.

Question 16

a. 

b. 

Question 17

a. **i.** The structure comprises of positive ions surrounded by delocalised electrons/electrons free to move within the structure.

ii. The structure comprises of positive ions and negative ions attracted by electrostatic forces throughout a lattice/network.

b. **i.** Delocalised electrons in magnesium are free to move through the structure and carry the current. In solid magnesium oxide, the charged particles/ions are held in place.

ii. The structure of magnesium metal can be deformed without destroying the forces of attraction between the ions and the electrons. Any disruption of the lattice/network in magnesium oxide will cause similar charged ions to be close together, causing repulsion and the destruction of the lattice.

Question 18

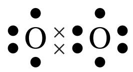
a. Aluminium will lose three electrons when it reacts, where sodium only loses one. Therefore, more energy is required when aluminium reacts.

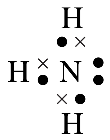
b. The diagram should show delocalised electrons and Al^{3+} ions in fixed positions.

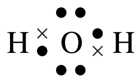
c. Aluminium can exist as different isotopes. The atomic mass is the average of the masses of these isotopes.

Question 19

a. 

b. 

c. 

d. 

Question 20

a. Allotropes are different physical forms of the same element.

b. Both consist of carbon atoms covalently bonded to other carbon atoms.

c. Diamond is a three-dimensional lattice where each carbon atom is covalently bonded to four other carbon atoms in a tetrahedral configuration, so strong bonding extends throughout the lattice.

Graphite consists of layers of carbon atoms where each is covalently bonded to three other carbon atoms, making strong layers. There are weaker dispersion forces between the layers. The one electron not involved in bonding is delocalised.

d. The delocalised electrons in graphite are free to move and conduct electricity. In diamond, each carbon atom is bonded with four other carbon atoms so there are no free electrons.

e. Because of the weak bonding between the layers of graphite, the layers can slide over one another and thus can slide onto a page.

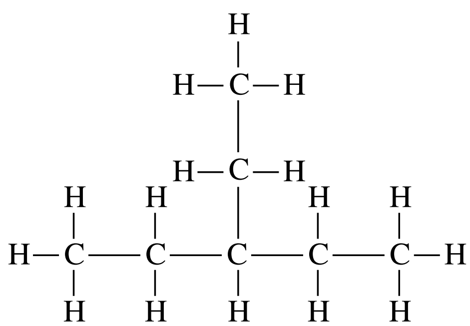
Question 21

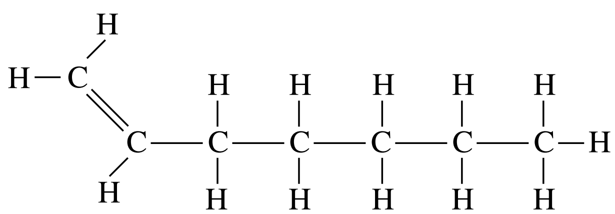
a. but-1-ene

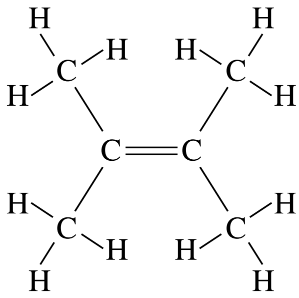
b. heptane

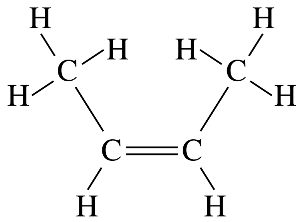
c. 2,2,3-trimethylpentane

Question 22

a. 

b. 

c. 

d. 

Question 23

a. $C_3H_8 + 5O_2 \rightarrow 3XO_2 + 4H_2O$

$2C_3H_8 + 7O_2 \rightarrow 6XO + 8H_2O$

$C_3H_8 + 2O_2 \rightarrow 3X + 4H_2O$

b. The amount of oxygen required to produce the carbon monoxide is less (3.5 moles of oxygen per 1 mole of propane) than that required to produce carbon dioxide (5 moles of oxygen per 1 mole of propane). Even less oxygen is required (2 moles of oxygen per 1 mole of propane) for the final reaction.

Question 24

a. fossilised organic material, mostly of plant origin

b. **i.** a mixture of hydrocarbons, mostly alkanes from C_1 to about C_{70}

ii. any correct formulas of two hydrocarbons, e.g. C_5H_{12}, C_6H_{14}

Question 25

a. Both processes are endothermic (because particles are being separated).

b. In the first process weak attractions between molecules are overcome. It is a physical change.

The second process is a chemical reaction, with strong chemical (covalent) bonds being broken. Therefore, more energy/heat is required.

Question 26

a. $n(\text{Al(NO}_3)_3) = \frac{m}{M} = \frac{30.5}{213} = 0.143$ mol

b. $n(\text{N}) = 3 \times 0.143 = 0.430$ mol

c. $\%\text{O} = 9 \times \frac{16}{213} \times 100 = 67.6\%$

Question 27

% by mass of oxygen in the compound = 100 – 59.7 = 40.3

In 100g of compound, m(O) = 40.3 g and m(X) = 59.7 g

Therefore, $n(\text{O}) = \frac{40.3}{16.0} = 2.52$

As ratio of X:O = 2:3, $n(\text{X}) = \frac{2}{3} \times 2.52 = 1.68$

Therefore 59.7 g of X is equivalent to 1.68 mol

And then $M(\text{X}) = \frac{m(\text{X})}{n(\text{X})} = \frac{59.7}{1.768} = 35.5 \text{ g mol}^{-1}$

The element must be Cl.

Question 28

%N in $KNO_3 = \frac{14.0}{101.1} \times 100 = 13.8\%$

So m(N) in 65.0 g KNO_3 = 9.00 g

%N in $(NH_4)_2SO_4 = \frac{28.0}{132.0} = 21.2\%$

So $m((NH_4)_2SO_4)$ that contains 9.00 g N $= \frac{9.00}{21.2} \times 100 = 42.5$ g

Question 29

a. $n(O_2) = (\frac{2}{1}) \times 1.34 = 2.68$ mol

b. $n(H_2O) = (\frac{2}{1}) \times 2.50 \times 10^{-2} = 5.00 \times 10^{-2}$ mol

$m(H_2O) = 5.00 \times 10^{-2} \times 18.016 = 0.900$ g

c. $n(C_8H_{18}) = (10 \frac{000}{114{,}224}) = 87.55$ mol

$n(H_2O) = (\frac{18}{2}) \times 87.55 = 787.9$ mol

$m(H_2O) = 787.9 \times 18.016 = 14\,195 \text{ g} = 14.2 \text{ kg}$

Question 30

a. butane

b. $n(C_4H_{10}) = \frac{n}{M} = \frac{220}{58.12} = 3.785 = 3.79$ mol

c. energy = 3.785 × 2859 = 10 822 = 10 800 kJ (to 3 significant figures)

or = 1.08×10^4 kJ

d. Energy transfers are never 100% efficient and some energy will be lost in the combustion reaction.

e. $n(O_2)$ required $= \frac{13}{2} \times n(C_4H_{10}) = \frac{13}{2} \times 3.785 = 24.6$ mol

$m(O_2) = n \times M = 24.6 \times 32.0 = 787$ g

f. butane $n(C_4H_{10}) = \frac{100}{58.12} = 1.72$

$n(CO_2) = \frac{8}{2} \times n(C_4H_{10}) = \frac{8}{2} \times 1.72 = 6.88$ mol

ethanol $n(C_2H_5OH) = \frac{100}{46.06} = 2.17$

$n(CO_2) = \frac{2}{1} \times n(C_2H_5OH) = \frac{2}{1} \times 2.171 = 4.342$ mol

Therefore, butane released (significantly) more carbon dioxide per 100 g of fuel than ethanol.

g. Increased carbon dioxide in the atmosphere contributes to the enhanced greenhouse effect, which has the potential to cause climate change.